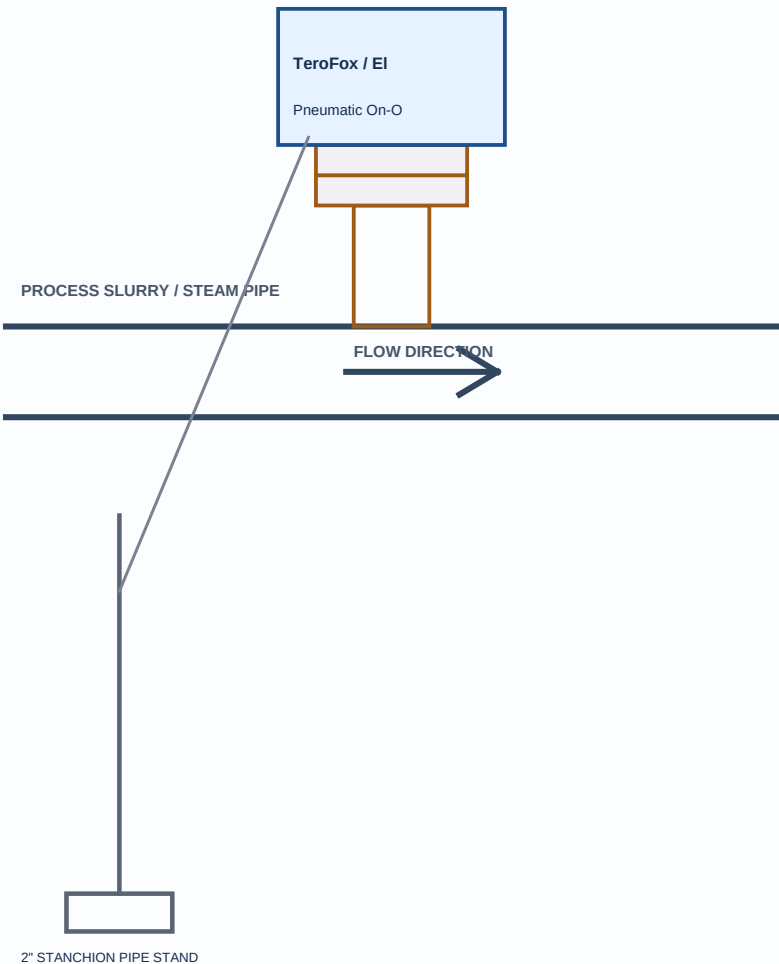


TEMPLATE TYP-20: PNEUMATIC ON-OFF VALVE

TYPICAL TAGS: XV-40101 / XV-60201

SERVICE: Automated Slurry Routing & Emergency Process Isolation | PLC I/O: 1756-OB32 (DO Solenoid) & 1756-IB32 (DI Open/Closed)

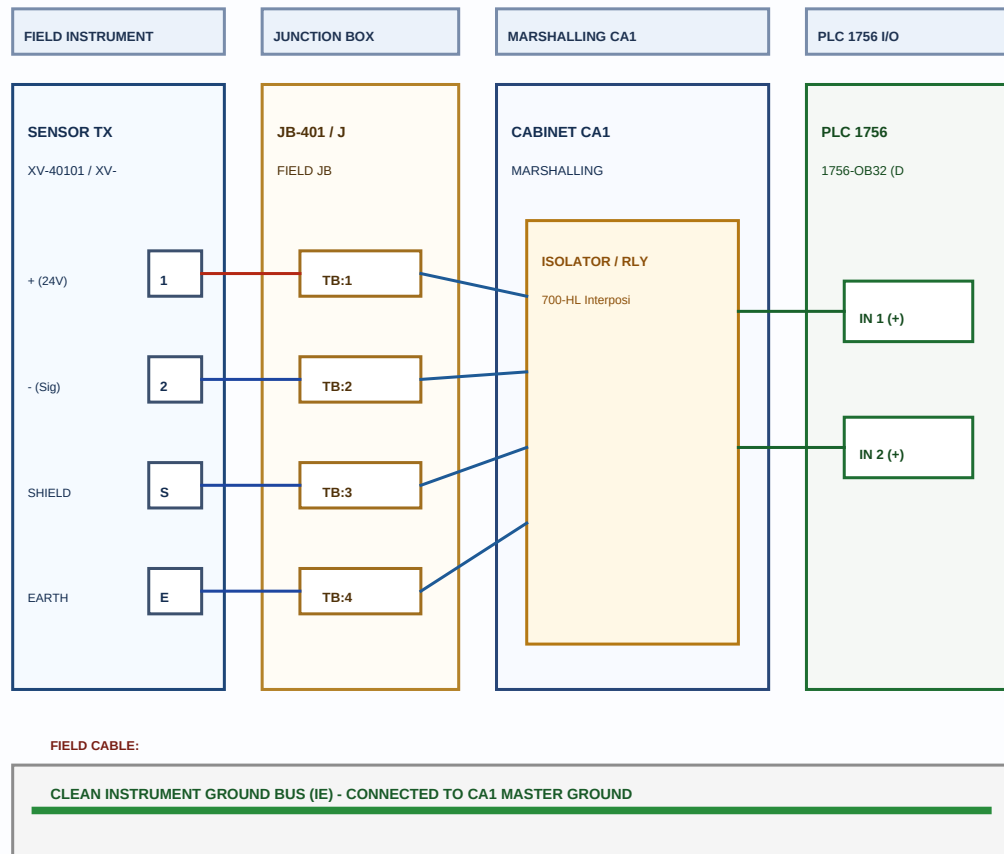
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	Flange ASME B16.5 Cl.150 RF...
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	Hard Anodized Aluminum Actua..
Hazardous Class:	ATEX II 2G Ex d IIC T6 Gb / ..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DO: 24VDC Energize to Open / DI: ZSO, ..
Power Supply:	24V DC from 1756-OB32 + 6 bar Clean In..
Field Cable Type:	3Cx1.5mm² (Solenoid) + 4Cx1.5mm² (Limi..
Destination JB:	JB-401 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	700-HL Interposing Relay (DO) + Fuses
PLC Module:	1756-OB32 (DO Solenoid) & 1756-IB32 (D..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: TEROFOX / EL-O-MATIC

MODEL: TF-20DFS Ball Valve + F-Series Actuator + Solenoid + APL-210N

SERVICE: Automated Slurry Routing & Emergency Process Isolation

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Pneumatic Ball Valve Assembly TeroFox TF-20DFS DN50 Cl.150		1 EA
2	Pneumatic Spring-Return Actuat El-O-Matic F-Series Rack & Pinio		1 EA
3	Namur Solenoid Valve 5/2-Way (Asco / SMC Low Power Solenoid		1 EA
4	Limit Switch Box APL-210N (2x Visual Yellow/Red Open/Close Bea		1 EA
5	Air Filter Regulator & Gauge S Modular AFR 1/4" w/ Auto-Drain		1 SET

ENGINEERING INSTALLATION NOTES

- Spring-return fail-safe action: Valve drives to CLOSED position on electrical power or air failure.
- Interposing relay 700-HL at CA1 cabinet isolates PLC 1756-OB32 output from field coil inrush.
- Air supply lines must use seamless SS316 1/4" or 3/8" tubing with Swagelok compression fittings.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTPNEUMATIC ON-OFF CONTROL VALVE HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-020
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
22 / 25